

Technical Document #1005: Retrieval Augmented Generation - Comprehensive Overview

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Query decomposition breaks complex queries into simpler sub-queries. It improves retrieval quality for multi-faceted questions.

Context window optimization selects the most relevant information to fit within the LLM's context limit. It maximizes information density.

Hybrid search combines keyword-based and semantic search. It leverages the strengths of both approaches for better retrieval quality.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Data-driven decision making has become essential for modern businesses. Organizations that leverage data effectively gain a significant competitive advantage.

Voice assistants like Alexa and Siri use speech recognition and natural language understanding. They have become integral parts of smart home ecosystems.

Federated learning trains models across decentralized data sources without sharing raw data. It enables privacy-preserving machine learning.

Key Takeaways:

- Hybrid search combines keyword-based and semantic search.
- Evaluation metrics for RAG include faithfulness, answer relevance, and context recall.
- Retrieval-Augmented Generation (RAG) combines information retrieval with text generation.